

Binary Responses

A response that can only be zero or one is the smallest departure from the normal linear model that still changes everything. Its mean is a probability, so it lives in an interval rather than on a line; its variance is determined by its mean, so there is no free dispersion parameter; and the exact distribution theory of Parts III and IV is gone. What remains is the likelihood, and the apparatus of [Chapter 34](#) applies word for word. This chapter is what that apparatus looks like pointed at the simplest possible response, and what that response does to it: odds ratios that survive sampling schemes no other model can tolerate, a maximum likelihood estimate that sometimes fails to exist, and a deviance that, for ungrouped data, carries no information about fit at all.

What you need. [Chapter 34](#) for the exponential dispersion family, the likelihood equations, iteratively reweighted least squares, the deviance and the asymptotics; [Chapter 7](#) for maximum likelihood and Fisher information (Theorem 7.3.1, Proposition 7.4.7); [Chapter 11](#) for the likelihood ratio test in the normal model (Theorem 11.4.1, Proposition 11.4.2); [Chapter 20](#) for residuals, leverage and influence; [Chapter 21](#) for weighted least squares; [Chapter 22](#) for empirical logits (Exercise (B2)); [Chapter 25](#) for the causal reading of a coefficient; [Chapter 27](#) and [Chapter 30](#) for penalized estimation; [Chapter 29](#) for AIC and cross-validation. Only [Chapter 34](#) is a prerequisite in the strict sense: every proof in this chapter that cites an earlier chapter cites that one. The rest are named because the chapter points at them — for a diagnostic, a penalty, a criterion or a reading of a coefficient — and a reader may follow the pointer when it arises.

35.-1.1 Roadmap

- [Logistic regression](#)
- [Probit and complementary log–log links](#)
- [Separation](#)
- [Goodness of fit](#)
- [Case–control sampling](#)
- [Summary and notes](#)